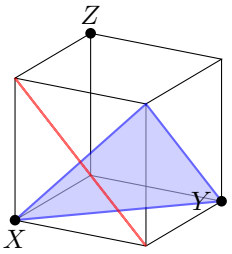
1 Exercise 1

consider the cube $C = (x, y, z) \in \mathbb{R}^3 | 0 \leq x \leq 1, 0 \leq y \leq 1, 0 \leq z \leq 1$ *in* $\mathbb{R}^3$. *Find the angle between the plane* P_1 *containing the 3 points:* $(0,1,0), (1,0,0), (1,1,1)$ *and the plane* P_2 *containing the 3 points:* $(0,0,0), (1,1,0), (1,0,1)$.



To find the angle between the two planes, we must firstly find the normal vector that the individual planes define and then find the angle between those two.

$$\vec{P_{1X}} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \qquad \vec{P_{1Y}} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \qquad \vec{P_{1Z}} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

Vectors for P_1

$$\vec{P_{1XY}} = \vec{P_{1X}} - \vec{P_{1Y}} = \begin{pmatrix} 0-1 \\ 1-0 \\ 0-0 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \qquad \vec{P_{1XZ}} = \vec{P_{1X}} - \vec{P_{1Z}} = \begin{pmatrix} 0-1 \\ 1-1 \\ 0-1 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ -1 \end{pmatrix}$$

The normal vector for plane $P1$ is the cross product of the two vectors in the plane.

$$\vec{N1} = \vec{P_{1XY}} \times \vec{P_{1XZ}} = \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix}$$

The procedure goes for the second set of points, to generate the set of vectors, the plane they sit on, and the normal vector that extends from it.

Points

$$\vec{P_{2X}} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \qquad \vec{P_{2Y}} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \qquad \vec{P_{2Z}} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

Vectors for $P2$

$$\vec{P_{2XY}} = \vec{P_{2X}} - \vec{P_{2Y}} = \begin{pmatrix} 0-1 \\ 0-1 \\ 0-0 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \\ 0 \end{pmatrix}, \qquad \vec{P_{2XZ}} = \vec{P_{2X}} - \vec{P_{2Z}} = \begin{pmatrix} 0-1 \\ 0-0 \\ 0-1 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ -1 \end{pmatrix}$$

Normal vector from plane $P2$

$$\vec{N2} = \vec{P_{2XY}} \times \vec{P_{2XZ}} = \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix}$$

The angle between these two vectors will then indicate the angle between the planes $P1$ and $P2$

$$\theta = cos^{-1}(\frac{N1 \cdot N2}{|N1||N2|}) \Rightarrow \frac{(-1)(1) + (-1)(-1) + (1)(-1)}{\sqrt{(-1)^2(-1)^2(1)^2} \cdot \sqrt{(1)^2(-1)^2(-1)^2}} \Rightarrow \frac{-1}{\sqrt{(3)^2}} \rightarrow = cos^{-1}(\frac{-1}{3}) * \frac{180}{\pi} \approx 109.492$$

2 exercise 2

Find the center of mass $(\bar{x}, \bar{y}, \bar{z})$, for the wire C shaped as a helix, with density $p = kz$, (k is a constant). The wire C has parametrization:

$$x(t) = 3cos(t), y(t) = 3sin(t), z(t) = 4t, 0 \leq t \leq \pi$$

The following equations define the center of mass:

- The curve length $s(t)$, as a function of t satisfies: $\frac{ds}{dt} = \sqrt{x'(t)^2 + y'(t)^2 + z'(t)^2}$
- $m = \int_c p(x, y, z)ds$
- $\bar{x} = \frac{1}{m}\int_c xp(x, y, z)ds, \bar{y} = \frac{1}{m}\int_c yp(x, y, z)ds, \bar{z} = \frac{1}{m}\int_c zp(x, y, z)ds$

This is a rather convoluted question, essentially asking, if we have a piece of wire, in this case a helix, meaning there is a one large curve, since it is bounded to $0 \leq t \leq \pi$, what is the position in space of the center of its mass. The parametrization describes how the *coil* will expand if we increase to. The first two components x and y are trigonometric functions, causing the direction to be circular around the z-axis, and the z component simply indicates that it increases linearly with t.

Firstly we need to calculate the change in t into changes on the actual physical wire. We're given the density p and mass is the product of density and lenght. So the curve length is calculated as:

$$\frac{ds}{dt} = \sqrt{x'(t)^2 + y'(t)^2 + z'(t)^2}$$

differentiating the functions

$$x'(t) = \frac{d}{dt}x(t) \Rightarrow -3sin(t)$$

$$y'(t) = \frac{d}{dt}y(t) = \frac{d}{dt}3cos(t)$$

$$z'(t) = \frac{d}{dt}z(t) = \frac{d}{dt}t(t)$$

into the curve length

$$\frac{ds}{dt} = \sqrt{(-3 \cdot sin(t))^2 + (3 \cdot cos(t))^2 + (4)^2}$$

rearrange, use distributive property and recognize the idiot formula $sin^2(t) + cos^2(t) = 1$

$$= \sqrt{9 + 1 + 16} = \sqrt{25} = 5$$

$$ds = 5dt$$

Relationship for density is given, with z given and k is some constant, it can we rewritten as:

$$p = k \cdot z$$

$$p = 4tk$$

Substituted into the mass relationship, with a change of variable from the $\frac{ds}{dt}$ relationship, so that mass is now in terms of t:

$$m = \int_0^{\pi} (4tk)(5)dt \Rightarrow 10k\pi^2$$

Since the lower bound is zero and are a single term with factors, it gets reduced to zero. Each of the coordinates for the center of mass can now be evaluated.

$$\bar{x} = \frac{1}{10k\pi^2}\int_0^{\pi} 3 \cdot cos(t) \cdot 4t \cdot k \cdot 5dt$$

$$= \frac{60k}{10k\pi^2}\int_0^{\pi} t \cdot cos(t)dt \Rightarrow \frac{60k}{10k\pi^2}[t \cdot sin(t) + cos(t)]_0^{\pi}$$

$$= \frac{6}{\pi^2} \cdot (-1) - \frac{6}{\pi^2} + cos(0) = -\frac{12}{\pi^2}$$

The same can be done for the two other coordinates:

$$\bar{y} = \frac{1}{10k\pi^2} \int_0^\pi 3 \cdot sin(t) \cdot 4t \cdot k \cdot 5dt$$

$$= \frac{60k}{10k\pi^2} \int_0^\pi t \cdot sin(t)dt \Rightarrow \frac{6}{\pi^2}[sin(t) - t \cdot cos(t)]_0^\pi \Rightarrow \frac{6}{\pi}$$

$$\bar{z} = \frac{1}{10k\pi^2} \int_0^\pi (4t) \cdot 4t \cdot k \cdot 5dt$$

$$= \frac{80k}{10k\pi^2} \int_0^\pi t^2 dt \Rightarrow \frac{8}{\pi^2}[\frac{t^3}{2}]_0^\pi \Rightarrow \frac{8\pi}{3}$$

Giving coordinates:

$$[\bar{x}, \bar{y}, \bar{z}] = [-\frac{12}{\pi^2}, \frac{6}{\pi}, \frac{8\pi}{3}]$$

3 exercise 3

Let F be a vector field on $\mathbb{R}^3$ given by:

$$F(x, y, z) = (0, 0, 1)$$

Using surface integrals, show that the flux through the upper hemisphere of the unit sphere:

$$S^2_+ = \{(x, y, z) \in \mathbb{R}^3 | x^2 + y^2 + z^2 = 1, z \geq 0\}$$

is the same as the flux through the unit disk in the xy-plane

$$D = \{(x, y, z) \in \mathbb{R}^3 | x^2 + y^2 \leq 1, z = 0\}$$

3.1 the unit sphere

we're given a vector field that essentially shows an upwards flow in in the Z direction, with a magnitude of 1.

$$F(x, y, z) = (0, 0, 1)$$

As for the hemisphere we're bounded for $z \geq 0$, therefore we can solve the hemisphere equation for z, essentially telling us the height z at any point in the sphere over (x, y) on the disc underneath.

$$x^2 + y^2 + z^2 = 1 \Rightarrow z = \sqrt{1 - x^2 - y^2}$$

this can be parameterized using x and y

$$r(x, y) = (x, y, \sqrt{1 - x^2 - y^2})$$

and x and y must lie within the unit disc

$$x^2 + y^2 = 1$$

We can then differentiate r with respect to x

$$r_x = \frac{\delta r}{\delta x} \Rightarrow r_x = (1, 0, \frac{-x}{\sqrt{1 - x^2 - y^2}})$$

and r with respect y

$$r_y = \frac{\delta r}{\delta x} \Rightarrow r_x = (0, 1, \frac{-y}{\sqrt{1 - x^2 - y^2}})$$

This lets us find the vector surface element $\mathbf{n}ds$, this is done by creaing the vector cross product

$$(r_x \times r_y) = (\frac{x}{\sqrt{1 - x^2 - y^2}}, \frac{y}{\sqrt{1 - x^2 - y^2}}, 1)$$

this leads to the expression

$$\mathbf{n}dS = (r_x \times r_y) = (\frac{x}{\sqrt{1 - x^2 - y^2}}, \frac{y}{\sqrt{1 - x^2 - y^2}}, 1)dxdy$$

This can be dottet with our vector field:

$$\mathbf{F} \cdot \mathbf{n}dS = (0, 0, 1) \cdot (\frac{x}{\sqrt{1 - x^2 - y^2}}, \frac{y}{\sqrt{1 - x^2 - y^2}}, 1)dxdy$$

which evaluates to

$$= 1dxdy$$

Now we can find the surface integral over the surface S, by using the vector field.

$$Q_s = \iint_S \mathbf{F} \cdot \mathbf{n} dS$$

The limits here can be derived:

$$x^2 + y^2 = 1 \Rightarrow x = \pm\sqrt{1 - y^2}$$

therefore y is bound within.

$$-1 \leq y \leq 1$$

and x is bound by

$$-\sqrt{1 - y^2} \leq x \leq \sqrt{1 - y^2}$$

Inserting these limits into the integral

$$Q_s = \int_{-1}^{1} \int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} 1 dx dy$$

These have been solved using MATHCAD The inner integration

$$\int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} 1 dx = 2\sqrt{1 - y^2}$$

And the outer aswell

$$\int_{-1}^{1} 2\sqrt{1 - y^2} dy = \pi$$

3.2 the unit disk

To calculate the flux through the disk.

Using the same field $F = (0, 0, 1)$, we need the normal vector to the surface of the disk in the same orientation as the vector field meaning $\mathbf{n} = (\hat{0}, 0, 1)$. The dot-product of these two vectors becomes $\mathbf{F} \cdot \hat{\mathbf{n}} = 1$, meaning they point in the same direction. This lets us create the integral of the disk flux.

$$\iint_D \mathbf{F} \cdot \hat{\mathbf{n}} dA = \iint_D 1 dA = \int_0^{2\pi} \int_0^1 r dr d\theta = \pi$$

This can also be shown when the hemisphere gets parameterized as $r(x, y) = (x, y, f(x, y))$ where $f(x, y) = \sqrt{1 - x^2 - y^2}$. We can then create to small vectors, representing small movement along the surface of the hemisphere $rx = (1, 0, fx), ry = (0, 1, fy)$. The cross product of those two, will then be perpendicular to them, as it'll become the normal vector to that tiny surface patch. The dot-product of this and the vector field F yields:

$$\mathbf{F} \cdot (rx \times ry) = 1$$

meaning that, that the vector picks out the z-component of the vector field. This leads to

$$\Phi_S = \iint_D 1 dA = \pi$$

4 exercise 4

let ϕ be a scalar field and F a vector field given by:

$$\phi(x, y, z) = xyz^2$$

and

$$F(x, y, z) = (x^2, y, z)$$

- Find $\nabla\phi$ and $\nabla \cdot F$ and $\nabla \cdot (\phi F)$
- try to deduce a general expression for $\nabla \cdot (\phi F)$ expressed by $\nabla\phi$ and $\nabla \cdot F$

Given the scalar field, the gradient ∇ can be found by differentiating once in each direction. The gradient of a scalar field is a vector field.

$$\nabla\phi = \begin{bmatrix} \frac{\partial\phi}{\partial x} \\ \frac{\partial\phi}{\partial y} \\ \frac{\partial\phi}{\partial z} \end{bmatrix} = \begin{bmatrix} yz^2 \\ xz^2 \\ 2xyz \end{bmatrix}$$

The next is finding the divergence of a vector field, which turns it into a scalar field.

$$\nabla F = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z} = 2x + 2$$